Examination Paper of Southeast University (B)

Course Name	Data Structures and Algorithms	Semester	18-19-2	Score	
Specialty	Software Engineering	Way of Examination	Open	Duration	120 min.
(Textbook, dictionary, class notes and exercise books are allowed)					

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姓名______ 学号______

I. Single choice (Each 4 points and total 40 points)

1 Given four algorithms with the following different time complexities, which one is the most inefficient? ( )

A. $O(n^{0.1})$ B. O(log(n))

C. O(nlog(n)) D. O(n)

2 Of the following data structures, which one is nonlinear?

A. Stack B. Queue C. Chain D. Binary Tree

3 Given a push-to-stack sequence "12345", which one is not the possible pop-from-stack sequence? ( )

A. 12354 B. 23541

C. 54321 D. 35412

4 If a binary tree with linked representation has n nodes, then there is ( ) empty pointers.

A. n-1 B. n+1 C. 2n-1 D.2n+1

5 In an undirected graph with 50 vertices, the maximum number of edges is ( ).

A. 1250 B. 1225 C. 2450 D.2500

6 Given a binary tree with **post-order** traversal sequence "BAECD" and **in-order** traversal sequence "BDACE", what is the **pre-order** traversal sequence of the tree? ( )

A. BDECA B. DBECA

C. DBCAE　　　　D. BCDEA

7 What is the maximum number of inner nodes of a B-tree of order 4 with height 5? (　　)

A.43　　B. 21　　C. 85　　D. 341

8 Assume there is a ordered list consisting of 200 data items, using binary search to find whether a special item is in it, the maximum comparisons is (　　) times.

A. 9　　B. 8　　C. 7　　D. 6

9 In the worst case, which one of the following sort algorithms has the least efficiency in time? (　　)

A. Heap sort　　B. Quick sort　　C. Merge sort　　D. Insertion sort

10 There is a Hash table with the size of 11. The Hash function is H(k)=3k mod 11. Every key in the sequence {32, 13, 49, 24, 38, 21, 5} is inserted into this table one by one according its Hash value. How many times of comparison are made when the key 5 is inserted.

A. 5　　B. 6　　C. 7　　D. 8

II. Answer the following questions (Each 10 points and total 40 points)

1 Given a weighted graph G=(V, E) with V={1, 2, 3, 4, 5, 6, 7}; E={(1,2) **3**, (1,3) **5**, (1,4) **8**, (2,5) **10**, (2,3) **6**, (3,4) **15**, (3,5) **12**, (3,6) **9**, (4,6) **4**, (4,7) **20**, (5,6) **18**, (6,7) **25**}, please use **Prim's Algorithm** to build a minimum-cost spanning tree T by adding edges one at a time.

2 Given a list (20, 15, 14, 18, 21, 36, 40, 10, 22), please describe the process that we use **Quick Sort** algorithm to sort it.

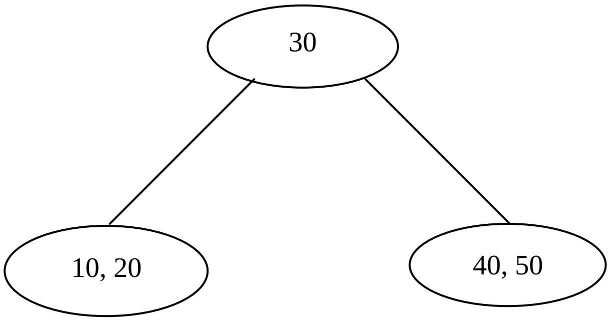
3 Let T be a initially empty **AVL tree**, draw the process of inserting 2, 11, 1, 13, 7, 17, 5, and 3 into T one by one, and write down the balance factors for each node and every rotating types if any.

4 For the following B-tree of order 3, please:

(1) draw the process of inserting15 and 60 into it.

(2) draw the process of deleting 10,30,20 from **the original B-tree**.

觉遵守考场纪律　如考试作弊　此答卷无效



III. Design algorithms (total 20 points)

1 Assume that there is a singular list (chain), in which the node structure is **<data, next>**, and the first node is pointed by **theFirst** pointer. Please design an algorithm to find the **k-th** (k is a positive integer) node **in the reversed order**. If it is successful, the algorithm outputs **data** of this node, and then returns 1; otherwise, the algorithm returns by 0.

The points you can get depend on the efficiency of the algorithm.

(1) Describe the idea of your algorithm in English or Chinese. (4 points)

(2) Write this algorithm in C++ language. Annotations in key points are required. (14 points)

(3) Analyze the time complexity and space complexity of your algorithm. (2 points)